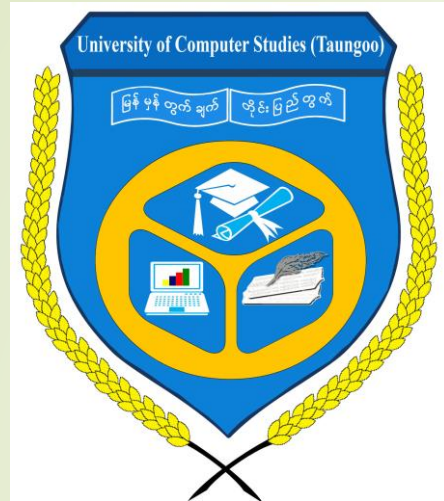




Introduction To Python

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Contents

- Modules
- NumPy
- Pandas
- SciPy
- Matplotlib

Modules

- A file containing a set of functions you want to include in your application.

Create a Module

- To create a module just **save** the code you want in a file with the file extension **.py**

Example:

```
def greeting(name):  
    print("Hello, " + name)
```

mymodule.py

```
import mymodule  
mymodule.greeting("Richard")
```

Module_Name

Function_Name

```
person={"name": "John",  
        "age":36, "country" : "Italy" }
```

mymodule.py

```
import mymodule  
a = mymodule.person["age"]  
print(a)
```

Modules

Re-naming a Module

- You can create an alias when you import a module, by using the **as** keyword.

Example:

```
def greeting(name):
```

```
    print("Hello, " + name)
```

mymodule.py

```
import mymodule as mx
```

```
mx.greeting("Richard")
```

Module_Name

Function_Name

Modules

Built-in Modules

- You can create an alias when you import a module, by using the **as** keyword.

Example:

```
import platform
x=platform.system( )
print(x)
y=dir(platform)
print(y)
```

Modules

Import From Module

- To import only parts from a module, by using the **from** keyword.

Example:

```
def greeting(name):  
    print("Hello, " + name)  
person={"name": "John",  
        "age":36, "country": "Italy"}
```

mymodule.py

```
from mymodule import person  
print(person["age"])
```

Numpy

Numpy (Numerical Python)

- Numpy is a **python library** used for working with arrays.
- It also has functions for working in domain of linear algebra, fourier transform, matrices.
- It was created in 2005 by Travis Oliphant. (open source project)

Create a NumPy ndarray Object

```
import numpy  
arr= numpy.array([1, 2, 3, 4, 5])  
print(arr)
```

```
import numpy as np  
arr= np.array([1, 2, 3, 4, 5])  
print(arr)
```


Numpy

- Use a tuple to create a NumPy array.

Dimension in Array

- A dimension in arrays is one level of array depth (nested arrays).

```
import numpy as np
a = np.array(42)
b = np.array([1, 2, 3, 4, 5])
c = np.array([[1, 2, 3], [4, 5, 6]])
d = np.array([[[1, 2, 3], [4, 5, 6]], [[1,2,3], [4, 5, 6]]])
print(a.ndim)      =>0
print(b.ndim)      =>1
print(c.ndim)      =>2
print(d.ndim)      =>3
```

```
import numpy as np
arr= np.array([1, 2, 3, 4, 5], ndmin=5)
print(arr)
print("number of dimension:", arr.ndim)
```



```
[[[[[1, 2, 3, 4, 5]]]]]
Number of dimension:5
```


Numpy

Array Indexing

```
import numpy as np
```

```
b = np.array([1, 2, 3, 4, 5])
```

```
print(b[0]+b[2]) =>4
```

```
c = np.array([[1, 2, 3, 7, 8], [4, 5, 6, 9, 10]])
```

```
print("5th element on 2nd row:", c[1,4]) =>10
```

```
d = np.array([[[1, 2, 3], [4, 5, 6]], [[7, 8, 9], [10, 11, 12]]])
```

```
print(d[0, 1, 2]) =>6
```

Numpy

Slicing Array

- Slicing in python means taking elements from one given index to another given index.
- **[start : end]**
- **[start : end : step]**

```
import numpy as np
b = np.array([1, 2, 3, 4, 5])
print(b[-3:-1])           => [5, 6]
print(b[ : : 2])          => [1,3, 5]
c = np.array([[1, 2, 3, 7, 8], [4, 5, 6, 9, 10]])
print(c[1, 1:4])          => [5, 6, 9]
print(c[0:2, 1:4])        => [ [2, 3, 7]
                               [5, 6, 9]]
```

Numpy

Data Types in Numpy

- i – integer
- b – boolean
- u – unsigned integer
- f – float
- c – complex float
- m – timedelta
- M – datetime
- O – object
- S – string
- U – unicode string
- V – fixed chunk of memory for other type (void)

```
import numpy as np  
arr= np.array([1, 2, 3, 4, 5])  
print(arr.dtype)           =>int64
```

```
import numpy as np  
arr= np.array([1, 2, 3, 4, 5], dtype='S')  
print(arr.dtype)
```

Numpy

Converting Array Data type	<code>astype()</code>	Create a copy of the array,, and allows you to specify the data type as a parameter
Numpy Array Copy	<code>np.copy()</code>	<i>owns</i> the data and any changes made to the copy will not affect original array
Numpy Array View	<code>np.view()</code>	change the original array
Numpy Array Join	<code>np.concatenate()</code>	Contents of two or more arrays in a single array
	<code>np.stack()</code>	Contents of two or more arrays in a single array
Numpy Array Split	<code>np.array_split()</code>	Reverse operation of joining
Numpy Searching Array	<code>np.where()</code> <code>np.sortedsearch()</code>	Search an array for a certain values, and return the indexes
Numpy Sort Array	<code>np.sort()</code>	Putting elements in an ordered sequence
Filtering Arrays	[True or False]	Filter an array using a boolean index list.

Numpy

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Random Numbers in Numpy

- **Random** : means something that can not be predicted logically.
- **Pesudo Random**: generate random numbers using a generation algorithm.
- **NumPy** offers the **random module** to work with **random numbers**.

```
from numpy import random
x=random.randint(100)      # generate a integer random from 0 to 100
print(x)                   =>67
y=random.rand()            # generate a float random from 0 to 1
print(y)                   =>0.000982
```

Numpy

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Random Numbers in Numpy

➤ Generate Random Array

```
from numpy import random
```

```
x=random.randint(100, size=(5))      #generate five integers random number array from 0 to 100
```

```
print(x)                =>[67, 12, 34, 78, 5]
```

```
y=random.randint(100, size=(3,5))    #generate 3 rows and 5 cols integer random number from 0 to 100
```

```
print(y)
```

➤ Generate Random Number From Array (choice())

```
from numpy import random
```

```
x=random.choice([3, 5, 7, 6])        #generate a random number from the given array
```

```
print(x)                =>6
```

```
y=random.choice([2, 4, 6, 8], size=(3,5))    #generate 3 rows and 5 cols random numbers from given array
```

```
print(y)
```


Pandas

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- **Pandas (Panel Data and Python Data Analysis)**
- Pandas is a **python library** and used for working with **data sets**.
- It has functions for analyzing, cleaning, exploring, and manipulating data.
- It was created by Wes McKinney in 2008.

```
import pandas as pd
mydataset = {'car':["BMW", "Volvo", "Ford"],
             'passings': [3, 7, 2] }
myvar = pd.DataFrame(mydataset)
print(myvar)
```



	cars	passings
0	BMW	3
1	Volvo	7
2	Ford	2

Pandas

➤ Pandas Series

- Pandas series is like a column in a table.

```
import pandas as pd
x = [1, 7, 2]
myvar = pd.Series(x)
print(myvar)
print(myvar[0]) ?
```

↓

```
0    1
1    7
2    2
dtype:int64
```

➤ Labels

- The values are labeled with their index number.

➤ Create Labels

- With the *index* argument, your own labels can be named.

```
import pandas as pd
x = [1, 7, 2]
myvar = pd.Series(x, index = ["a", "b", "c"])
print(myvar)
print(myvar[a])?
```

➡

```
a    1
b    7
c    2
dtype:int64
```

Pandas

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➤ Pandas Read CSV(comma separated files)

➤ Pandas series is like a column in a table.

```
import pandas as pd
df = pd.read_csv('data.csv')
print(df)                                # will only return the first 5 rows, and the last 5 rows
print(df.to_string())                    # to_string(): to print the entire DataFrame
```

➤ Pandas Read JSON

➤ Pandas series is like a column in a table.

```
import pandas as pd
df = pd.read_json('data.json')
print(df.to_string())                    # to_string(): to print the entire DataFrame
```

Pandas

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➤ Pandas - Analyzing DataFrames

Viewing the Data	head()	Returns the headers and a specified number of rows, starting from the top. (default parameter = 5rows)
	tail()	Returns the headers and a specified number of rows, starting from the bottom. (default parameter = 5rows)
Info About the Data	info()	Gives more information about the data set.

```
import pandas as pd
df = pd.read_csv('data.csv')
print(df.head( ))
print(df.tail( ))
print(df.info( ))
```

Pandas

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➡ Pandas – Cleaning Data

1. **Empty cells** : give a wrong result when data is analyzed.

```
import pandas as pd
df = pd.read_csv('data.csv')
df.dropna(inplace = True)      # return new data frame that remove all rows with Null values
print(df.to_string( ))
```

```
import pandas as pd
df = pd.read_csv('data.csv')
df.fillna(130, inplace = True)  # replace empty cells with a value
df["Calories"].fillna(130, inplace = True ) #replace null values in the "Calories" columns with the number 130
```

Pandas

➤ Pandas – Cleaning Data

Replacing Using Mean, Median, or Mode

```
import pandas as pd
df = pd.read_csv('data.csv')
x = df["Calories"].mean( )
x = df["Calories"].median( )
x = df["Calories"].mode( )[0]
df["Calories"].fillna(x, inplace = True )    #replace any empty value with respective values
```

Pandas

➤ Pandas – Cleaning Data

2. **Wrong Format** : make difficult or even impossible, to analyze data.

```
import pandas as pd
df = pd.read_csv('data.csv')
df["Date"] = pd.to_datetime(df["Date"])
df.dropna(subset=['Date'], inplace = True)    # remove rows with a NULL value in the "Date" column
print(df.to_string( ))
```

Pandas

➡ Pandas – Cleaning Data

3. **Wrong Data** : it can just be wrong, like if someone registered “199” instead of “1.99”.

```
import pandas as pd
```

```
df = pd.read_csv('data.csv')
```

```
df.loc[7, 'Duration'] = 45
```

```
for x in df.index:
```

```
    if df.loc[x, "Duration"] > 120:
```

```
        df.loc[x, "Duration"] = 120
```

```
for x in df.index:
```

```
    if df.loc[x, "Duration"] > 120:
```

```
        df.drop(x, inplace = True)
```

#Replacing Values (set “Duration” = 45 in row 7)

#If the value is higher than 120, set it to 120

#Removing Rows (Delete rows where “Duration” is higher than 120)

Pandas

➡ Pandas – Cleaning Data

4. Removing Duplicates: registered more than one time.

```
import pandas as pd
df = pd.read_csv('data.csv')
print(df.duplicated( ))           #Returns True for every row that is a duplicate, otherwise False
df.drop_duplicates(inplace = True) #To remove duplicates
```

Pandas

- **Pandas – Plotting**
- To create diagram, **plot()** method is used in pandas.
- To specify a scatter plot with the **kind** argument: (**kind='scatter'**)
- To specify a histogram, also use **kind** argument : (**kind='hist'**)

```
import pandas as pd  
df = pd.read_csv('data.csv')  
df.plot()  
df.plot(kind='scatter', x='Duration', y='Calories')  
df['Duration'].plot(kind='hist')
```

Exercises

- <https://pynative.com/python-operators-and-expression-quiz/>
- <https://pynative.com/python-variables-and-data-types-quiz/>
- <https://pynative.com/basic-python-quiz-for-beginners/>
- <https://pynative.com/python/quizzes/>
- <https://pynative.com/python-basic-exercise-for-beginners/>
- https://www.w3schools.com/python/python_functions.asp

Thank You